

Supplementary Material for “WEAVE: Breaking the Identity Shortcut in Lightweight Style Transfer with Wavelets”

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Abstract—This supplement documents the evaluation protocol, the resolved training configuration, the internal stopping probe, per-epoch curves for every seed and retrained ablation, sensitivity sweeps, reference-pool and ArtFID audits, structure-preservation measurements, the high-frequency route probes that motivated the final architecture, and the frozen SD1.5 plug-in. Section, table, and figure numbers prefixed with “S” refer to this document; all others refer to the main paper.

S1. EVALUATION PROTOCOL

D5-512. Distinct5-WikiArt at 512 px contains Early Renaissance, Impressionism, Minimalism, Rococo, and Ukiyo-e, with 3,600 training and 150 test images per style. Each of 30 test sources per style is sent to all five targets, giving 750 ordered requests; the 150 same-style requests are kept as calibration rows. Training uses a frozen prototype pairing cache built from training latents only.

P2A-256. A 256 px five-domain grid over photo, Monet, Van Gogh, Cézanne, and Hayao, evaluated with the same ordered logic. Several external baselines only provide the 600 off-diagonal requests on this board; their scores are averaged over the images they provide, and image counts are recorded with every score.

R5-WikiArt. Five randomly drawn WikiArt families (Cubism, Expressionism, Pop Art, Romanticism, Symbolism) at 512 px with the same 750-request layout.

Metrics. DINO-S is the mean over requests of the maximum cosine similarity between the DINOv2-small CLS feature of the output and the CLS features of up to 30 held-out references of the target style (the source is excluded from the pool). CLIP-S is the CLIP ViT-B/32 cosine similarity to the target-style prototype. DINO-C is the DINOv2-small CLS cosine between output and source. LPIPS uses the AlexNet backbone. Images are resized to 224 and center-cropped for DINO; metric post-processing is disabled.

IDT and TGT. For source x and target style s , $y_{\text{IDT}} = x$ and $y_{\text{TGT}} = r_s$, where r_s is the first image in the deterministic held-out reference order of style s , reused for every source sent to s . TGT depends on the requested style but not on the source. At test time WEAVE uses the same r_s as its style reference, so TGT is exactly the output of copying the reference; Section S6 verifies that the DINO-S margin of WEAVE does not rely on r_s being in the evaluation pool. We read the bracket

$$\begin{aligned} M_{\text{style}}(y) &> M_{\text{style}}(y_{\text{IDT}}), \\ \text{LPIPS}(y, x) &< \text{LPIPS}(y_{\text{TGT}}, x), \\ \text{DINO} - C(y, x) &> \text{DINO} - C(y_{\text{TGT}}, x) \end{aligned} \quad (1)$$

TABLE S1
RESOLVED CONFIGURATION OF THE REPORTED D5-512 RUN.

Field	Value
Decomposition	one-level Haar on the VAE latent
Endpoint	source-aligned LL, $\alpha = 0.3$; target LH/HL/HH
Band weights	LL 0.3, LH 1.0, HL 1.0, HH no head
Path	$t \sim \mathcal{U}[0, 1]$, $\sigma = 0.02$ perturbation
Target-HF route	pooled LH/HL codes, gate init 0.18
Optimizer	AdamW, lr $2 \times 10^{-4} \rightarrow 10^{-5}$ cosine, wd 10^{-4}
Batch / clipping	96 / 1.0, bf16 AMP, TF32, channels-last
Epoch budget	15, internal stop (fired at epoch 4)
Probe	4 fixed latents at $t = 0.5$, after each epoch
Inference	8 Euler steps, AdaLN $\beta = 2.0$, $\eta = 0.1$
Seed / GPU	42 / one RTX 3060 (12 GB)

inequality by inequality and never combine it into a weighted score. TGT reaches DINO-S 1.000 by construction because r_s belongs to the reference pool. Using each of the first five deterministic exemplars as r_s moves TGT LPIPS within 0.752–0.783 and TGT DINO-C within 0.189–0.246 on D5; StyleAligned (LPIPS 0.869) lies beyond every one of these anchors.

S2. IMPLEMENTATION

Table S1 lists the resolved configuration of the reported D5 model. The frozen SD1.5 EMA VAE (84M parameters, latent scale 0.18215) is used only to encode training latents once and to decode outputs. The velocity network has four main residual blocks (plus two high-resolution and two decoder blocks) at width 64 and time embedding 256; with the orientation-specific target-HF route it has 1,037,087 trainable parameters (873,680 without the route).

Training log. Each epoch has 52 optimizer steps. Losses for epochs 1–4 are 1.3631, 1.3318, 1.3199, and 1.2499, with epoch times 25.72, 19.23, 18.94, and 18.90 s (82.80 s in total, including the probe; the first epoch warms caches and kernels). Generation-only inference for the 750 D5 requests takes 126.0 s (168 ms per image) including VAE decoding and excluding metric extraction.

Internal-dynamics probe. For the fixed probe batch the model runs in evaluation mode, performs one backward pass for the LL loss and one for the summed LH/HL loss, and aggregates gradient norms over shared input, attention, AdaLN, and feed-forward modules; no optimizer step is taken and random states are restored. Table S2 gives the trace. Epoch 4 is the first epoch with a positive gate change and a shared

TABLE S2
INTERNAL PROBE TRACE OF THE REPORTED RUN (SEED 42).

Epoch	Gate	Δ_{gate}	LL/HF (shared)	route/HF	route/head
1	0.1731	0.0000	1.4727	0.8323	0.1816
2	0.1723	-0.0008	1.4328	0.6133	0.1907
3	0.1720	-0.0003	1.0868	0.4984	0.2111
4	0.1757	+0.0037	0.1634	0.2370	0.4561

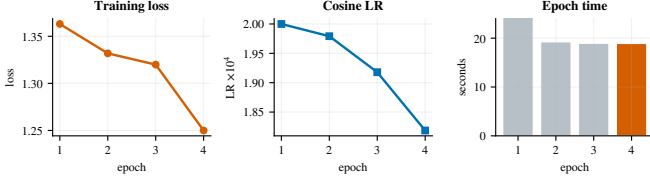


Fig. S1. Training dynamics of the reported D5-512 run: loss, cosine learning rate, and epoch time. Training stops at epoch 4.

LL/HF ratio that contracts to at most 0.65 of the previous epoch ($0.1634/1.0868 = 0.15$). The rule is relative; an earlier absolute crossing-at-one rule was scale-sensitive and did not fire for seed 7 within 15 epochs. Fig. S1 shows loss, learning rate, and epoch time.

S3. SPECTRAL DIAGNOSIS AND PROPOSITION 1

Measurement. For 5,000 cross-style D5 latent pairs (z_c, z_s) we compute the one-level orthonormal Haar transform of the direct displacement $u = z_s - z_c$ and average the band energies $\|u_b\|^2$. Because the transform is orthonormal, $\|v - u\|^2 = \sum_b \|v_b - u_b\|^2$, so at initialization ($v \approx 0$) the gradient of the flow-matching loss with respect to the band- b output is $-2u_b$ and its energy share equals the displacement share. The mean energies are 4.241 (LL), 0.628 (LH), 0.687 (HL), and 0.546 (HH), i.e., shares of 69.5, 10.3, 11.3, and 8.9%. Style separability is the ratio of between-style to within-style variance of per-image band statistics: 0.117 (LL), 0.186 (LH), 0.171 (HL), and 0.560 (HH).

Consequence of Proposition 1. The source-anchored endpoint leaves the high-frequency displacement unchanged and, by the bound of Proposition 1 in the main paper, scales the LL energy by at most $\alpha^2 = 0.09$ for every pair, hence also on average. The LL share therefore becomes at most $0.09 \cdot 4.241 / (0.09 \cdot 4.241 + 0.628 + 0.687 + 0.546) = 0.170$. The bound is attained only when the reference LL is a positive affine copy of the source LL, so the actual share is lower. In the implementation, AdaIN uses per-sample, per-channel spatial statistics with a 10^{-6} floor on the standard deviation, which does not affect the bound in practice. The derivation is reproduced by `tools/make_figures.py`.

S4. PER-EPOCH CURVES AND STOPPING REGRET

Every checkpoint of every run was evaluated offline on the full D5 board to validate the online, metric-free stopping rule (Table S3). The external metric never influences training. For the main configuration the rule selects the DINO-S maximum

TABLE S3
STOPPING RULE VERSUS DINO-S ORACLE ON D5-512. e_{int} : EPOCH CHOSEN BY THE INTERNAL RULE; e^* : BEST DINO-S EPOCH AMONG EVALUATED CHECKPOINTS.

Run	epochs	e_{int}	e^*	DINO-S(e^*)	regret
WEAVE, seed 42	4	4	4	0.4917	0
WEAVE, seed 7	15	4 ^a	4	0.4910	0
WEAVE, seed 123	3	3	3	0.4862	0
$\lambda_{LL} = 1.0$	15	5	3	0.4910	6.8×10^{-5}
direct endpoint	15	5	13	0.4894	0.0039
learned HH head	15	4	4	0.4930	0

^arelative rule; the superseded absolute rule did not fire for this seed.

for seeds 42 and 123 and, for seed 7, the epoch that is the maximum of its full 15-epoch curve. For the retrained ablations, which were run for 15 epochs with stopping disabled, the rule’s choice differs from the oracle: by 6.8×10^{-5} DINO-S for $\lambda_{LL} = 1.0$, but by 0.0039 for the direct endpoint, whose peak arrives at epoch 13. The main-paper ablation table therefore compares every retrained variant at WEAVE’s four-epoch budget, which shares seed and learning-rate schedule with the main run, and lists each variant’s best DINO-S within 15 epochs separately.

For seed 42, (DINO-S, CLIP-S, LPIPS, DINO-C) is (0.4857, 0.7119, 0.2409, 0.8230) after epoch 1, (0.4887, 0.7158, 0.2625, 0.7978) after epoch 2, (0.4913, 0.7176, 0.2645, 0.7853) after epoch 3, and (0.4917, 0.7127, 0.2595, 0.8104) after epoch 4. The main table reports a re-evaluation of the same epoch-4 checkpoint (0.4918/0.7128/0.2595/0.8102); the difference is below 2×10^{-4} on every metric. After epoch 4, DINO-S declines while the regression loss keeps improving, so the transition is not ordinary loss-plateau stopping. Across the three seeds at their selected epochs, (DINO-S, CLIP-S, LPIPS, DINO-C) is $(0.4897 \pm 0.0030, 0.7137 \pm 0.0008, 0.2605 \pm 0.0059, 0.8073 \pm 0.0031)$, mean $\pm$ s.d.; we use these spreads as the noise yardstick for the ablations.

S5. SENSITIVITY SWEEPS

Table S4 reports three one-factor sweeps. The AdaIN scale sweep changes only inference on the reported checkpoint and is fully matched. The λ_{LL} and α sweeps retrain the model at a fixed budget of four to five epochs, whereas the 0.3/0.3 reference point comes from the internally stopped main run. WEAVE needs no fine tuning of λ_{LL} : across $\lambda_{LL} \in [0.1, 0.5]$ the retrained DINO-S stays within 0.4847–0.4857, while DINO-C decreases and CLIP-S increases monotonically with λ_{LL} , giving a smooth style–content dial. Along α , DINO-C and LPIPS both fall, so the two content proxies capture different aspects of fidelity, which is why the sandwich checks both. AdaIN scale is flat between 1.0 and 1.5 and changes sharply at 2.0, where DINO-S, DINO-C, and LPIPS improve while CLIP-S drops slightly; values above 2.0 were not tested on this checkpoint (on the earlier base model, scale 2.5 collapses content: DINO-C 0.259).

TABLE S4
SENSITIVITY SWEEPS ON D5-512 (DINO-S/DINO-C/CLIP-S/LPIPS).

Setting	DINO-S	DINO-C	CLIP-S	LPIPS
<i>Stepwise AdaIN scale β (same checkpoint)</i>				
1.0	0.4831	0.8024	0.7165	0.2863
1.25	0.4838	0.8026	0.7167	0.2865
1.5	0.4844	0.8031	0.7168	0.2864
2.0	0.4920	0.8099	0.7126	0.2593
<i>LL loss weight λ_{LL} (retrained, fixed budget)</i>				
0.1	0.4856	0.8206	0.7110	0.2585
0.15	0.4857	0.8202	0.7111	0.2589
0.2	0.4856	0.8185	0.7119	0.2598
0.25	0.4853	0.8178	0.7121	0.2595
0.35	0.4850	0.8151	0.7135	0.2591
0.4	0.4847	0.8141	0.7136	0.2575
0.45	0.4850	0.8122	0.7141	0.2577
0.5	0.4848	0.8106	0.7143	0.2573
<i>LL blend α in Eq. (1) (retrained, fixed budget)</i>				
0.1	0.4859	0.8243	0.7096	0.2714
0.2	0.4864	0.8215	0.7108	0.2651
0.4	0.4845	0.8090	0.7148	0.2557
0.5	0.4843	0.8016	0.7152	0.2552
reference (main run)	0.4918	0.8102	0.7128	0.2595

TABLE S5
DINO-S UNDER REFERENCE-POOL RESAMPLING (1,000 DRAWS, D5-512).

m	WEAVE	IDT	margin (95% interval)	positive
8	0.345	0.315	0.0298 [0.0204, 0.0394]	100% / 68.9%
16	0.378	0.346	0.0319 [0.0225, 0.0380]	100% / 69.2%
30	0.403	0.370	0.0334 (full pool)	—

“positive”: fraction of draws with positive board margin / of individual requests.

S6. REFERENCE-POOL STABILITY

Without regenerating images, we draw for each target style one size- m subset of its 30 held-out references without replacement, reuse it for all 150 requests to that style, and recompute DINO-S for WEAVE and IDT; this is repeated 1,000 times (Table S5). Absolute DINO-S grows with m because it is a maximum over the pool, but the board-mean margin over IDT is positive in every repetition. Because WEAVE conditions on the first held-out image of each style, this test also controls for that image being in the pool: a random 8-image pool contains it with probability 8/30, and the margin with such pools (0.030) is close to the full-pool margin (0.033). Roughly 69% of individual requests have a positive margin.

S7. ARTFID AUDIT

ArtFID is computed per target style as $(1 + \text{FID})(1 + \text{LPIPS})$ and averaged over targets on a single 750-request source manifest (Table S6). Only methods whose outputs match that manifest are compared; Z-STAR and StyleAligned outputs were produced from a different source list and are excluded, and Seedream (API) covers 720 requests and is shown for reference. IDT obtains the best ArtFID and also the best raw FID, so ArtFID alone cannot distinguish successful

TABLE S6
TARGET-POOLED ARTFID ON D5-512 (LOWER IS BETTER).

Method	FID	LPIPS	ArtFID
IDT	215.5	0.000	216.5
WEAVE	230.5	0.283	295.3
SaMam	231.9	0.283	297.3
Seedream 4.5 (720 req.)	217.4	0.422	311.0
TGT (30 random draws)	305.3	0.787	545.7 $\pm$ 56.1

transfer from no transfer; this motivates the IDT reference. WEAVE and SaMam are close on ArtFID, but WEAVE clears IDT on CLIP-S while SaMam does not (main Table I; its CLIP-S is below IDT on D5 and R5).

S8. STRUCTURE PRESERVATION

As a geometry check independent of DINO and LPIPS, we compare MiDaS (DPT) depth maps and Canny edge maps of source and output on all 750 D5 requests. WEAVE has a lower depth MSE than SaMam (0.0321 vs. 0.0406; median 0.0182 vs. 0.0216), while SaMam has a higher edge IoU (0.159 vs. 0.140), consistent with WEAVE re-texturing edges while keeping scene depth. On the 15 requests for which StyleAligned outputs match the manifest, StyleAligned has depth MSE 0.129 and edge IoU 0.004, against 0.043 and 0.368 for WEAVE on the same requests: three times the depth error and almost no surviving edges.

S9. HIGH-FREQUENCY ROUTE PROBES

Before the final architecture was fixed, we diagnosed why the base model moved style weakly. The structure-aligned target already requests target high frequencies, but in the base model the target latent shapes the supervision without entering the velocity predictor as an image-specific condition. We therefore fine-tuned alternative target-HF routes from one base checkpoint (AdaIN scale 1.5, D5 board; diagnostic only, not mixed into the main table), see Table S7 and Fig. S2. Raw spatial target maps raise style but leak target layout (DINO-C 0.404, LPIPS 0.538). Pooled subband codes keep content while raising style; zeroing them at inference lowers both DINO-S and DINO-C, so the route is causal. Scaling the route ($1.25\times$, $1.5\times$) trades content for off-diagonal style without improving the balanced point, and a direction auxiliary improves a decomposition probe but hurts image metrics. The final model uses orientation-specific pooled codes for LH and HL only.

S10. ADDITIONAL COMPARISONS

Fig. S3 repeats the sandwich plot of the main paper for every benchmark, and Fig. S4 shows the training and inference cost of the learned methods. WEAVE lies in the valid region on all three benchmarks; at 512px (D5, R5) no baseline lies above and to the right of it in any style-content pair.

Unseen families. The five held-out WikiArt families are disjoint from D5; all methods use their D5 checkpoints and the same 750-request layout (Table III of the main paper).

TABLE S7
TARGET-HF ROUTE PROBES (FINE-TUNES OF ONE BASE CHECKPOINT,
D5-512).

Route	DINO-S	DINO-C	LPIPS	CLIP-S
HF delta, 15 epochs	0.4825	0.7913	0.2991	0.7180
raw spatial target HF	0.4901	0.4043	0.5382	0.7483
pooled subband codes	0.4886	0.7981	0.2966	0.7209
codes zeroed at inference	0.4858	0.7888	0.3010	0.7205
codes $\times 1.5$	0.4875	0.7798	0.3054	0.7211
subband + texture stats	0.4884	0.7988	0.2960	0.7194
without style memory	0.4849	0.7948	0.2943	0.7167
direction auxiliary	0.4862	0.7939	0.2974	0.7189

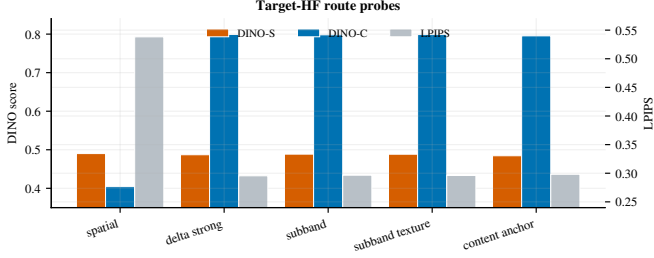


Fig. S2. Target-HF route probes. Spatial injection raises style but collapses content; pooled subband routes keep content at nearly the same style level.

Frozen SD1.5 plug-in. The control is an SD1.5 image-to-image pipeline with an empty prompt, strength 0.5, and 20 DDIM steps. The plug-in applies a one-level Haar transform to the output latent, aligns the channel moments of LH, HL, and HH to the target-style latent at scale 0.8, leaves LL unchanged, and inverts the transform. On all 750 requests (including same-style rows), DINO-S changes from 0.4525 to 0.4613 and CLIP-S from 0.7476 to 0.7594, while LPIPS stays at 0.3586/0.3588 and DINO-C moves from 0.6583 to 0.6539. The operation costs 4.9 ms per image (720 timed requests). The frequency-domain operator is thus portable: it upgrades a billion-parameter editor as a drop-in step.

S11. ETHICS AND REPRODUCIBILITY

Style transfer can be misused to produce deceptive images or misattribute authorship. WEAVE operates at the level of style families (e.g., Impressionism) and does not model or label individual artists, although inference uses target-style reference images. All data are public WikiArt artworks without personal data, and the VAE is the public SD1.5 EMA VAE. Code, the selected checkpoint, and the 750-request manifests will be released upon publication. Reported training time starts after one-off latent caching; inference time includes VAE decoding.

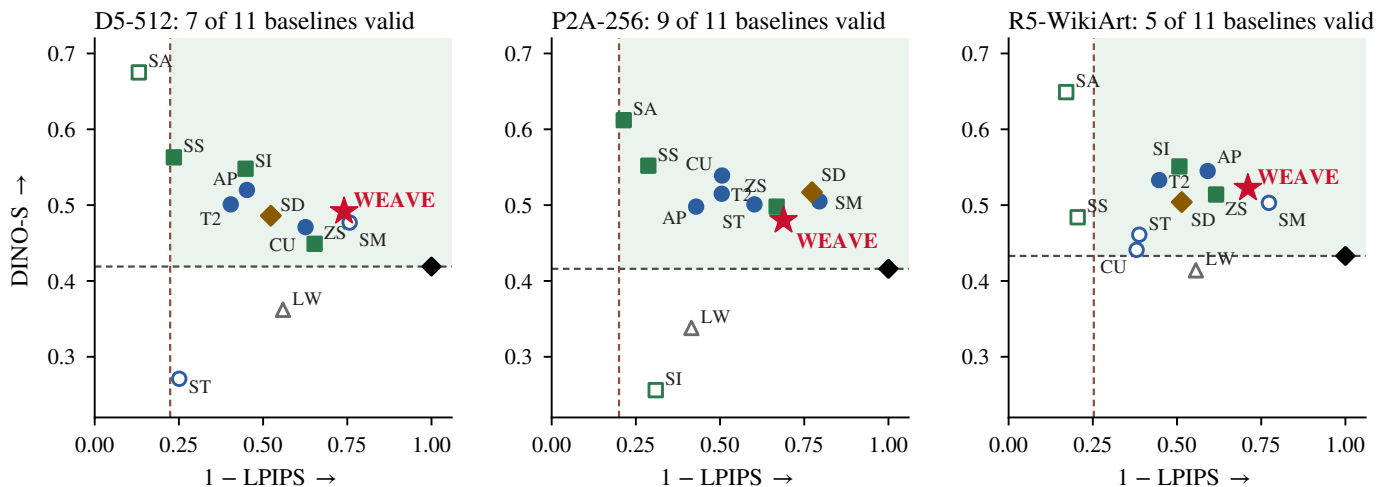


Fig. S3. The IDT-TGT sandwich on all benchmarks (DINO-S versus 1-LPIPS; dashed lines: IDT style and TGT content; shaded: valid region on these two axes; hollow markers fail at least one of the four tests). SA: StyleAligned, ZS: Z-STAR, SS: StyleShot, SI: StyleID, CU: CUT, ST: SaMST, SM: SaMam, T2: StyTr², AP: AesPA-Net, SD: Seedream 4.5, LW: Latent-WCT; black diamond: IDT.

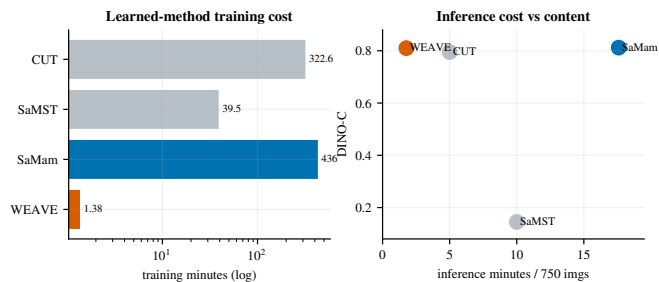


Fig. S4. Training minutes (left, log scale) and inference minutes versus DINO-C (right) for the learned methods.